

QCD-like theories at NNLO

Daniil Krichevskiy

University of Stavanger
Department of Mathematics and Physics

Joint work with Helena Kolečová and Suchita Kulkarni
(see JHEP 05 (2026) 042),

and with Johan Bijnens

Journal club, 20.05.2026, University of Lund

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Beyond-the-Standard-Model scenarios propose new QCD-like, strongly interacting sectors, e.g:

- Composite pionic dark matter, e.g. Strongly Interacting Massive Particles (**SIMPs**),
- Composite Higgs models.

At low energies, we assume the non-zero condensate which breaks the flavor symmetry and Chiral perturbation theory is used to describe the dynamics of pseudo-Goldstone bosons from spontaneous symmetry breaking.

$$\langle \bar{q}q \rangle \neq 0 \quad \begin{array}{c} G \\ \downarrow \\ H \end{array}$$

Flavour symmetries

Depending on the representation of the N_F fermionic flavors the chiral symmetry breaking pattern differs:

Representation of the fermions	Complex	Pseudoreal	Real
Example: fundamental of	$SU(N_c)$	$Sp(N_c)$	$O(N_c)$
Chiral symmetry breaking pattern	$SU(N_F) \times SU(N_F)/SU(N_F)$	$SU(2N_F)/Sp(2N_F)$	$SU(2N_F)/SO(2N_F)$
Number of pions	$N_F^2 - 1$	$(2N_F + 1)(N_F - 1)$	$N_F(2N_F + 1) - 1$

UV theory

N_F flavors of left-handed fermions and N_F flavors of right-handed fermions in the fundamental representation of $Sp(N_c)$ or $SO(N_c)$:

$$\mathcal{L} = \bar{q}_{Li} i\gamma^\mu D_\mu q_{Li} + \bar{q}_{Ri} i\gamma^\mu D_\mu q_{Ri} - \bar{q}_{Ri} \mathcal{M}_{ij} q_{Lj} - \bar{q}_{Li} \mathcal{M}_{ij}^\dagger q_{Rj}.$$

(Pseudo-)real representation: $\exists$ a unitary matrix ϵ such that

$$-T^{a*} = \epsilon^{-1} T^a \epsilon, \quad \epsilon^T = \eta \epsilon, \quad \begin{cases} \eta = 1, & \text{real fermions} \\ \eta = -1, & \text{pseudoreal fermions} \end{cases}$$

With the definition

$$\hat{q} = \begin{pmatrix} q_R \\ \epsilon C \bar{q}_L^T \end{pmatrix} \Big\}^{2N_F}, \quad \hat{\mathcal{M}} = \begin{pmatrix} 0 & \eta \mathcal{M} \\ \mathcal{M}^T & 0 \end{pmatrix}, \quad C = i\gamma^2 \gamma^0$$

we can rewrite the Lagrangian as

$$\mathcal{L} = \bar{\hat{q}} i\gamma^\mu D_\mu \hat{q} - \frac{\eta}{2} \hat{q}^T \hat{\mathcal{M}}^\dagger \epsilon^* C \hat{q} - \frac{1}{2} \bar{\hat{q}} C \epsilon \hat{\mathcal{M}} \bar{\hat{q}}^T.$$

$$\mathcal{L} = \bar{q} i \gamma^\mu D_\mu \hat{q} - \frac{\eta}{2} \hat{q}^T \hat{\mathcal{M}}^\dagger \epsilon^* C \hat{q} - \frac{1}{2} \bar{q} C \epsilon \hat{\mathcal{M}} \bar{q}^T$$

- if $\hat{\mathcal{M}} \rightarrow 0$, global $U(2N_F)$ symmetry,
- axial anomaly breaks it down to $SU(2N_F)$,
- Chiral condensate breaks the symmetry spontaneously:

$$\sum_{i=1}^{N_F} \langle \bar{q}_i q_i \rangle = \langle \frac{1}{2} \hat{q}^T J \epsilon^* C \hat{q} + \frac{1}{2} \bar{q} C \epsilon J \bar{q}^T \rangle \neq 0, \quad J = \begin{pmatrix} 0 & \eta \mathbf{1}_{N_F} \\ \mathbf{1}_{N_F} & 0 \end{pmatrix},$$

The condensate is invariant under $H \subset SU(2N_F)$ such that

$$J = h^T J h, \quad h \in H \subset G = SU(2N_F), \quad \begin{cases} H = SO(2N_F), & \eta = 1 \\ H = Sp(2N_F), & \eta = -1 \end{cases}.$$

Generators split into broken (X^a) and unbroken (Q^a) parts:

$$J Q^a + Q^{aT} J = 0, \quad J X^a - X^{aT} J = 0.$$

External vector as sources

Non-zero chiral condensate $\Rightarrow$ spontaneous symmetry breaking.
Broken currents $A_\mu^a = \bar{\hat{q}}\gamma_\mu X^a \hat{q}$ have divergence

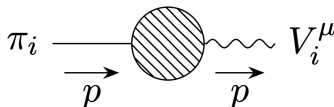
$$\partial^\mu A_\mu^a \neq 0.$$

To generate the correlation functions with the insertions of j_μ^a we enhance the Lagrangian with external vector fields

$$\mathcal{L} = \bar{\hat{q}} i \gamma^\mu D_\mu \hat{q} - \frac{\eta}{2} \bar{\hat{q}}^T \hat{\mathcal{M}}^\dagger \epsilon^* C \hat{q} - \frac{1}{2} \bar{\hat{q}} C \epsilon \hat{\mathcal{M}} \bar{\hat{q}}^T + \bar{\hat{q}} \gamma^\mu V_\mu \hat{q},$$

where $V_\mu = V_\mu^a X^a$.

$$\langle 0 | A^a(0)_\mu | \pi^a(p) \rangle = i\sqrt{2} p_\mu F_{\pi,a} \Rightarrow$$



Effective theory

The EFT is written in terms of the matrix (CCWZ construction)

$$u = e^{\frac{i}{\sqrt{2}F}\pi^a X^a} \in G/H, \quad F - \text{pion decay constant.}$$

Using u we can define, the object valued in the unbroken part of G :

$$u_\mu = i \left(u^\dagger (\partial_\mu - iV_\mu) u - u (\partial_\mu + iJV_\mu^T J^T) u^\dagger \right) \xrightarrow{G} hu_\mu h^\dagger.$$

Further building blocks including quark masses read:

$$\chi_\pm = u^\dagger \chi J^T u^\dagger \pm u J \chi^\dagger u \xrightarrow{G} h\chi_\pm h^\dagger, \quad \chi = 2B_0 \hat{M}.$$

The χ PT is an expansion in small pion momentum

$$\frac{p^2}{(4\pi F)^2} \sim \frac{M^2}{(4\pi F)^2} < 1.$$

We count $\partial \sim \mathcal{O}(p)$ and $m_q \sim \mathcal{O}(p^2)$.

LO and NLO Lagrangians

Chiral Lagrangian consists of terms systemized by number of derivatives and quark mass insertions:

$$\mathcal{L}_2 = \frac{F^2}{4} \langle u_\mu u^\mu + \chi_+ \rangle,$$

$$\begin{aligned} \mathcal{L}_4 = & L_0 \langle u^\mu u^\nu u_\mu u_\nu \rangle + L_1 \langle u^\mu u_\mu \rangle \langle u^\nu u_\nu \rangle + L_2 \langle u^\mu u^\nu \rangle \langle u_\mu u_\nu \rangle + L_3 \langle u^\mu u_\mu u^\nu u_\nu \rangle \\ & + L_4 \langle u^\mu u_\mu \rangle \langle \chi_+ \rangle + L_5 \langle u^\mu u_\mu \chi_+ \rangle + L_6 \langle \chi_+ \rangle^2 + L_7 \langle \chi_- \rangle^2 \\ & + \frac{1}{2} L_8 \langle \chi_+^2 + \chi_-^2 \rangle - i L_9 \langle f_{+\mu\nu} u^\mu u^\nu \rangle + \frac{1}{4} L_{10} \langle \not{f}_+ - \not{f}_- \rangle \\ & + H_1 \langle l_{\mu\nu} l^{\mu\nu} + r_{\mu\nu} r^{\mu\nu} \rangle + H_2 \langle \chi \chi^\dagger \rangle, \end{aligned}$$

$\mathcal{L}_6 \supset 115$ operators with LECs K_i .

The LECs, L_i and K_i , are renormalized to cancel the loop divergencies $\frac{1}{\varepsilon}$ and $\frac{1}{\varepsilon^2}$.

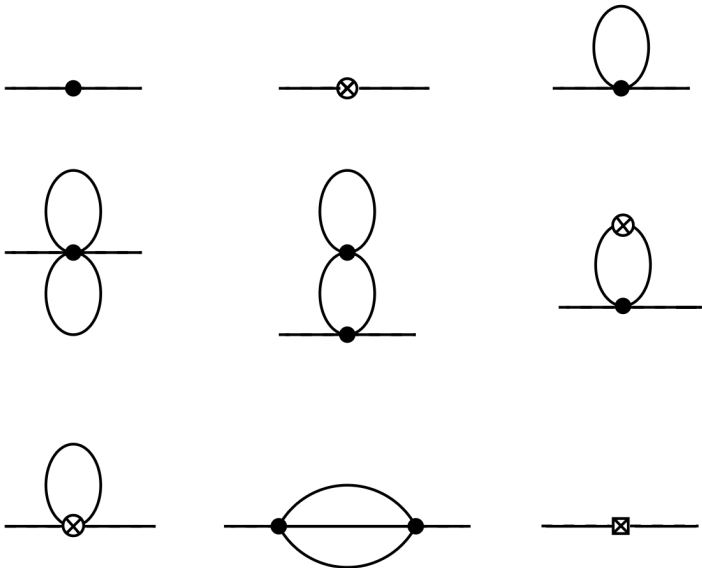
Higher-order corrections are important for the BSM scenarios with relatively large pion mass, M_π . These corrections are important for

- phenomenology of SIMP dark matter (pointed out by Hansen, Langæble and Sannino, **1507.01590**)
- interpolation between the limit of massless quarks and the finite-mass regime where lattice simulations can be performed (relevant for the composite Higgs models)

For theories with N_F (pseudo-)real fermions the NLO and NNLO results are presented in papers by Bijnens and Lu **0910.5424** and **1102.0172**.

We extend the $N_F = 2$ NLO and NNLO (with the exception of scattering amplitude) results by assuming non-degenerate quark masses (specially relevant for SIMP phenomenology).

Example: masses for $SU(4)/Sp(4)$ at NNLO



Example: masses for $SU(4)/Sp(4)$ at NNLO

LO contribution:

$$M^2 \equiv M_{\text{LO},k}^2 = B_0(m_u + m_d), \quad k = 1 \dots 5.$$

NLO contribution:

$$M_{\text{NLO}}^2 = \frac{M^2}{F^2} (a_M \bar{A}(M^2) + b_M M^2 + \alpha_M R^2 M^2)$$

NNLO contribution

$$M_{\text{NNLO}}^2 = \frac{M^2}{F^4} \left(c_M \bar{A}(M^2)^2 + M^2 \bar{A}(M^2) (d_M + \pi_{16} e_M) \right. \\ \left. + M^4 (f_M + g_M \pi_{16} + h_M \pi_{16}^2) + M^2 R^2 (M^2 \beta_M + M^2 \pi_{16} \gamma_M + \bar{A}(M^2) \delta_M) \right).$$

$$R = \frac{m_u - m_d}{m_d + m_u}, \quad \pi_{16} = \frac{1}{16\pi^2}, \quad \bar{A}(M^2) = -\pi_{16} M^2 \log \frac{M^2}{\mu^2}$$

Example: masses for $SU(4)/Sp(4)$ at NNLO

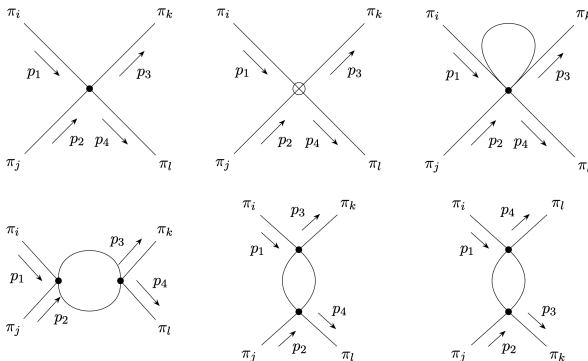
Pion species	All
a_M	$-\frac{3}{4}$
b_M	$16L_8^r + 64L_6^r - 8L_5^r - 32L_4^r$
c_M	$\frac{69}{32}$
d_M	$4L_8^r + 16L_6^r - 2L_5^r - 8L_4^r + 22L_3^r + 40L_2^r + 88L_1^r - 2L_0^r$
e_M	$\frac{3}{2}$
f_M	$r_{M,0}^r - 128L_5^r L_8^r - 512L_5^r L_6^r + 64(L_5^r)^2 - 512L_4^r L_8^r - 2048L_4^r L_6^r + 512L_4^r L_5^r + 1024(L_4^r)^2$
g_M	$12L_8^r + 48L_6^r - 6L_5^r - 24L_4^r + L_3^r + 12L_2^r + 4L_1^r + 5L_0^r$
h_M	$\frac{5}{6}$
Pion species	$\pi_1, \pi_2, \pi_4, \pi_5$
α_M	0
β_M	$r_{M,1}^r$
γ_M	$4L_8^r + 16L_7^r$
δ_M	$-4L_8^r - 16L_7^r$
Pion species	π_3
α_M	$16L_r^8 + 64L_r^7$
β_M	$r_{M,2}^r - 128L_r^5 L_8^r - 512L_r^5 L_r^7 - 512L_4^r L_8^r - 2048L_4^r L_r^7$
γ_M	$-4L_8^r - 16L_r^7$
δ_M	$20L_r^8 + 80L_r^7$

2 $\rightarrow$ 2 scattering amplitude at NLO

$$\mathcal{M}^{ij \rightarrow kl} = \langle \pi_k(p_3) \pi_l(p_4) | \pi_i(p_1) \pi_j(p_2) \rangle$$

Less symmetric structure compared to the degenerate case, especially for $SU(4)/SO(4)$ theory!

See Mathematica functions publicly available at [Zotero repository](#).



We obtained the results for other observables, to be presented in the paper:

- decay constants,

$$\langle 0 | A^a(0)_\mu | \pi^a(p) \rangle = i\sqrt{2}p_\mu F_{\pi,a}, \quad a = 1 \dots N_\pi,$$

- condensates,

$$\langle \bar{u}u \rangle = \frac{\partial V_{\text{eff, min}}}{\partial m_u}, \quad \langle \bar{d}d \rangle = \frac{\partial V_{\text{eff, min}}}{\partial m_d}.$$

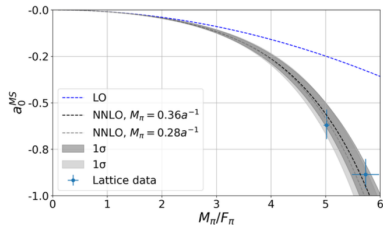
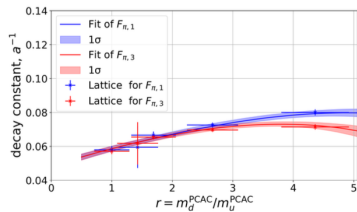
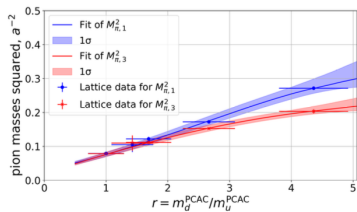
Fitting the lattice data

The LECs for $Sp(N_c = 4)$, $N_F = 2$ theory were not yet known. We perform a fit using the derived formulas of the lattice data:

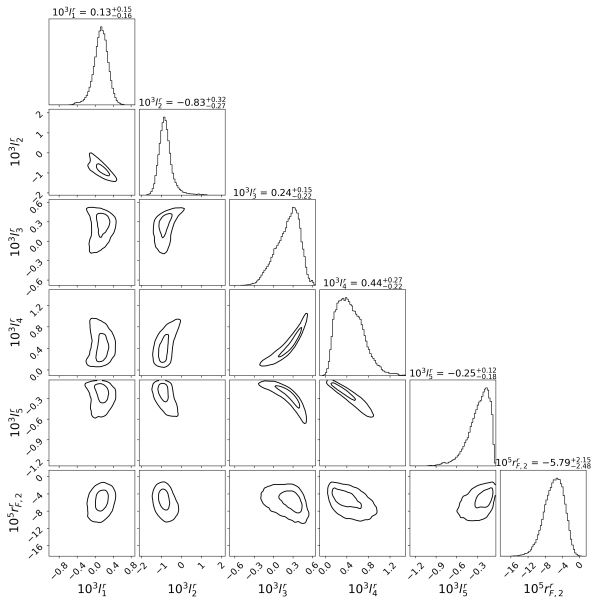
- masses and decay constants for the non-degenerate quarks: Kulkarni et al., **2202.05191v2**. The lattice show splitting in the decay constants which can not be captured by the NLO formulas.
- scattering amplitude for the mass-degenerate quarks in the channel without s-channel vector mesons: Dengler, Maas, Zierler, **2405.06506v2**.

The papers use similar lattice ensembles. The continuum limit is not performed in both papers.

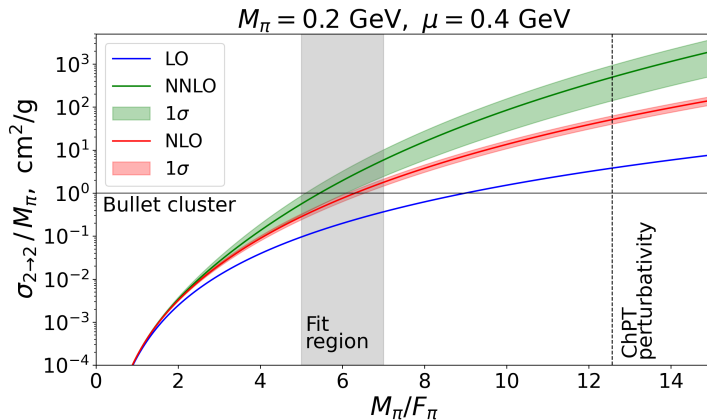
Results of the fit



Fitted LECs



Self-scattering cross section

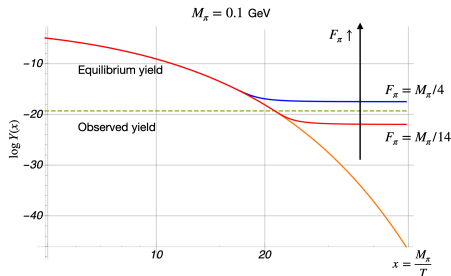
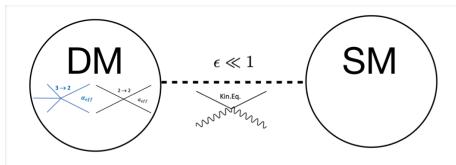


SIMP dark matter

Consider a scenario in which the relic abundance is set by $3 \rightarrow 2$ self-interaction (Hochberg et al., **1402.5143v2**).

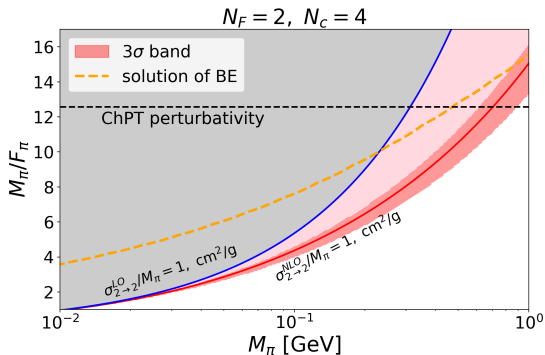
In χ PT there is 5-point interaction due to a topological **WZW term** (which gives $K^+ K^- \rightarrow \pi^+ \pi^- \pi^0$ in QCD):

$$\mathcal{L}_{\text{WZW}} \supset \frac{N_c}{240\pi^2 F_\pi^5} \epsilon^{\mu\nu\rho\sigma} \langle \pi \partial_\mu \pi \partial_\nu \pi \partial_\rho \pi \partial_\sigma \pi \rangle$$



Reproduced from 1402.5143

SIMP validity



The result should be taken with care: vector mesons are light and should be taken into account in the EFT (Bernreuther et al., **2311.17157**)
 $\Rightarrow$ ongoing project.

- NLO and NNLO formulas for masses, condensates, decay constants for theory with $N_F = 2$ real and pseudoreal non-degenerate fermions are derived ,
- first estimate of the NLO LECs for $Sp(N_c = 4)$ theory using the lattice data,
- confirmation of the importance of the NLO calculations for the dark matter theories with relatively high M_π/F_π , with refined computation using LECs fitted from the lattice.

Thank you for attention!